# Math 205A - Fall 2014
# Homework #3
# Solutions

**Problem 1:** **(i)** Show that step functions and continuous functions are dense in $L^p[0,1], 1 \le p < \infty$. Is this true for $L^\infty$?
**(ii)** Show that if $f$ is integrable in $E$, then $\lim_{t\to 0} \int_E |f(x+t) - f(x)|dx = 0$.

**(i)** A step function is a function of the form $\sum_1^n c_i \chi_{A_i}$, where $A_i$'s are finitely many disjoint interpals.
Let $\mathcal{S}$ and $\mathcal{C}$ be the vector spaces of step functions and of continuous functions in $[0,1]$ respectively. (Note that finite linear combinations of step functions are again step functions, and the same holds true for continuous functions.) Want to show that $L^p = \bar{\mathcal{S}}$ and $L^p = \bar{\mathcal{C}}$, where closure is taken w.r.t. the $L^p$ norm.

$\underline{L^p \subset \bar{\mathcal{S}}}$ : First, $\chi_A \in \mathcal{S}$ for $A \subset [0,1]$ measurable: given $\epsilon$, $A \subset U$ for some open $U$, with $m(U - A) < \epsilon^p/2$. $U$ is a countable union of disjoint intervals $I_i$. Pick finitely many of those to enssure $m(U - \cap_1^n I_i) \le \epsilon^p/2$, so $m(A - \cup_1^n I_i) \le \epsilon^p$. Then $||\chi_A - \sum_1^n \chi_{U_i}||_p = m(A - \cup_1^n U_i)^{1/p} \le \epsilon$. Hence, $\chi_A$ can be approximated by step functions.
The same is true for simple functions, because if $f = \sum_1^\infty c_i \chi_{A_i}$ for disjoint $A_i$'s and $c_i \neq 0$, we can pick finitely many of those to ensure $||f - \sum_1^n c_i \chi_{A_i}||_p < \epsilon/2$, and then approximate each $\chi_{A_i}$ by the step function $s_i$ such that $||f - s_i||_p < 2^{-(i+1)} c_i^{-1} \epsilon$. $s = \sum_1^n s_i$ is again a step function, and $||f - s||_p < \epsilon$.
So simple functions are included in $\mathcal{S}$, therefore so are $L^p$ functions.

$\underline{L^p \subset \bar{\mathcal{C}}}$ : Exactly the same reasoning, up to an interpolation argument, works, because the step functions themselves can be approximated by continuous functions (in the $L^p$ norm). Indeed, given $\epsilon$ and an interval $[a,b] \subset [0,1]$, the function $f$ that equals 0 on $[a,b]^c$, 1 on some $[a',b'] \Subset [a,b]$ with $m([a,b]\backslash[a',b']) < \epsilon^p$, and linear interpolation in between, satisfies $||\chi_{[a,b]} - f||_p < \epsilon$. So continuous functions are dense in the step functions, and hence, $L^p$.

Step functions are <u>not</u> dense in $L^\infty$: let $f = \sum_{n=2}^\infty \chi_{[\frac{1}{2n+1}, \frac{1}{2n}]}$. $f \in L^\infty$, but $||f - s||_\infty \ge 1/2$ for any step function $s$.
Continuous functions are <u>not</u> dense in $L^\infty$, because $[0,1]$ is compact, and uniform ($L^\infty$) limits of uniformly continuous functions are continuous.

**(ii)** Note that the problem makes sense only if $f$ is defined on $E' := E + (-\delta, \delta)$ and all the $t$'s in consideration are $|t| < \delta$, so let's extend $f$ by 0 outside $E$. $f$ is integrable on $E'$, hence in $L^1(E')$. Step functions are dense there, so given $\epsilon$, pick $s$ a step function on $E'$, such that $||f - s||_{L^1(E')} < \epsilon$. Since $f = 0$ on $E'\backslash E$, we can take w.l.o.g. $s = 0$ there too. Because

$$\int_E |f(x+t) - f(x)| \le \int_E |f(x+t) - s(x+t)| + \int_E |s(x+t) - s(x)| + \int_E |f(x) - s(x)| < 2\epsilon + \int_E |s(x+t) - s(x)|,$$

it's enough to prove the statement for the step functions $s = \sum_1^n c_i \chi_{[a_i,b_i]}$ in $E$ ($f \in L^1$ implies $s \in L^1$, so all the intervals must be bounded). $|s(\cdot + t) - s(\cdot)|$ are dominated by $2\sum_1^n |c_i| \chi_{[a_i - \delta, b_i + \delta]}$, which is integrable, and they converge pointwise to 0 a.e.,so by dominated convergence $\int_E |s(x+t) - s(x)| \to 0$ as $t \to 0$. □

**Problem 2:** Show that if $\mu(E) < \infty$ and $f_n \to f$ a.e., then the followings are equivalent:
(i) $f_n$ are uniformly integrable, (ii) $\int |f_n - f| \to 0$, (iii) $\int |f_n| \to \int |f|$.

We will assume in this problem that $f$ is integrable, otherwise one can come up with counterexamples for the equivalence (for example $f_n = n\chi_{[0,1/n]} + x\chi_{(1/n,0)} \to 1/x = f$ a.e., and $\int_{[0,1]} |f_n| \to \int_{[0,1]} |f| = \infty$, but $f_n$ are not uniformly integrable).
**(i) $\implies$ (ii):** Let $\epsilon > 0$. From uniform integrability there exists an $\alpha_\epsilon$ such that $\int\limits_{\{|f_n| \ge \alpha_\epsilon\}} |f_n| < \epsilon/4$. Now pick $\delta < \frac{\epsilon}{4\alpha_\epsilon}$ such that $\mu(A) \le \delta \implies \int_A |f| \le \epsilon/4$. From Egorov's theorem, exists a measurable set $E_\epsilon \subset E$ such that

$\mu(E\backslash E_\epsilon) \le \min\{\epsilon/4, \delta\}$, and $f_n \to f$ uniformly in $E_\epsilon$. This implies that exists some $N_\epsilon$ such that for all $n > N_\epsilon$, $\int\limits_{E_\epsilon} |f_n - f| < \epsilon/4$, and therefore

$$\begin{aligned}\int\limits_E |f_n - f| &\le \int\limits_{E_\epsilon} |f_n - f| + \int\limits_{E\backslash E_\epsilon} |f_n| + \int\limits_{E\backslash E_\epsilon} |f| \\ &\le \tfrac{\epsilon}{4} + \int\limits_{\{|f_n|\ge\alpha_\epsilon\}} |f_n| + \int\limits_{\{|f_n|<\alpha_\epsilon\}\cap\{E\backslash E_\epsilon\}} |f_n| + \tfrac{\epsilon}{4} \\ &\le \tfrac{\epsilon}{2=4} + \alpha_\epsilon \mu(E\backslash E_\epsilon) + \tfrac{\epsilon}{4} + \tfrac{\epsilon}{4} \le \tfrac{3\epsilon}{4} + \alpha_\epsilon \cdot \delta \le \epsilon.\end{aligned}$$

This proves that $\lim \int |f_n - f| \le \epsilon$ for all $\epsilon$, hence (ii).
**(ii) $\implies$ (iii):** $\limsup \int |f_n| \le \lim \int (|f_n - f| + |f|) = \int |f|$. On the other hand, $\int |f| \le \int |f_n - f| + \int |f_n|$, so $\int |f| \le \liminf \int |f_n - f| + \liminf \int |f_n| = \liminf \int |f_n|$. Hence, $\lim \int |f_n - f|$ exists and equals $\int |f|$.
**(iii) $\implies$ (i):** Suppose not. Then there exists an $\epsilon > 0$, $\alpha_k \to \infty$ and a sequence of $n_k$'s such that $\int_{|f_{n_k}|\ge\alpha_k} |f_{n_k}| \ge \epsilon$. On the other hand, $g_k = |f_{n_k}|\chi_{\{|f_{n_k}|\ge\alpha_k\}} \to 0$ a.e. ($\chi_{\{|f_{n_k}|\ge\alpha_k\}} \to 0$ because $\int |f_n| \le \int |f| + \delta$ after some $N$). $g_k \le |f_{n_k}|$ and $\int |f_{n_k}| \to \int |f|$, so by problem 3, $\int g_k \to 0$, contradicting $\int g_k = \int_{|f_{n_k}|\ge\alpha_k} |f_{n_k}| \ge \epsilon$ □

**Problem 3:** **(i)** Show that if $|f_n| \le g \in L^1(\Omega)$, then $f_n$ are uniformly integrable in $\Omega$. Does there exist a uniformly integrable family $\{f_n\}$ with no integrable $g$ such that $|f_n| < g$?
**(ii)** Let $f_k$ and $g_k$ be $\mu$-measurable, such that $f_k \to f$ $\mu$-a.e., $g_k \to g$ $\mu$-a.e., $|f_k| \le g_k$ and $\int g_k \to \int g$. Show that $\int f_k \to \int f$.

**(i)** $\int_{|f_n|\ge\alpha} |f_n| \le \int_{g\ge\alpha} g$ since $|f_n| \le g$, and $\lim_{\alpha\to\infty} \int_{g\ge\alpha} g = 0$ since $g \ge 0$ is integrable.
*Yes.* $f_n = \chi_{[n,n+1]}$ on $\mathbb{R}$ are uniformly integrable (consider $\alpha > 1$), and if $f_n \le g$, then $g \ge 1$ on $(0, \infty)$, which is not integrable.
**(ii)** Note that here we need to assume that $g$ is integrable, otherwise the result is not true.
$|f_k| \le g_k$ implies $g_k - f_k \ge 0$ and $g_k + f_k \ge 0$. Apply Fatou's lemma to both:
$\int (g - f) \le \liminf \int (g_k - f_k) = \int g - \limsup \int f_k \implies \limsup \int f_k \le \int f$.
$\int (f + g) \le \liminf \int (f_k + g_k) = \liminf \int f + \int g \implies \liminf \int f_k \le \int f$.
Combining these gives the result. □

**Problem 4:** **(i)** Show that any increasing function is a sum of an absolutely continuous and a singular function.
**(ii)** Does there exist a strictly increasing singular function?

**(i)** Let $f$ be a monotone function. $f'$ exists a.e., so let $g(x) = \int_0^x f'$, and $h = f - g$. Then $g$ is absolutely continuous, and $h$ is singular.
**(ii)** *Yes.* Consider the strictly increasing function $f(x) = \sum_{q_n\in\mathbb{Q}} 2^{-n}\chi_{(q_n,\infty)}$ from HW2, and let $h$ be its singular part. $h$ is increasing because $h(y) - h(x) = f(y) - f(x) - \int_x^y f' \ge 0$ for $y > x$. If $h$ wasn't strictly increasing, then it would be constant on some interval $[x, y]$, hence continuous there, so $f = g + h$ would also be continuous on $[x, y]$, contradicting the discontinuity of $f$ on a dense subset of $[0, 1]$. □

**Problem 5:** Construct an absolutely continuous strictly increasing function on $[0, 1]$ such that $g' = 0$ on a set of positive measure.

Consider the set $E$ from HW1 with $0 < m(E \cap I) < |I|$ for all intervals $I$ in [0.1], and let $f(x) = \int_0^x \chi_E$. $f$ is an indefinite integral, hence *absolutely continuous.* It is *strictly increasing* since for $y > x$, $f(y) - f(x) = m(E \cap [x, y]) > 0$. This in turn implies that $f' = \chi_E$, so $f' = 0$ on $E^c$ with $m(E^c) > 0$. □